NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-23648/19B
4 April 1995
SUPERSEDING
MIL-T-23648/19A
16 October 1972

PERFORMANCE SPECIFICATION

RESISTOR, THERMAL (THERMISTOR),
HERMETICALLY SEALED, POSITIVE TEMPERATURE COEFFICIENT
STYLE RTH42

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RTH42 thermistors. This style is available in resistance ratio E only. Terminal leads type S and W are available. Resistance tolerance versus temperature characteristics G, J, and K are applicable through the maximum temperature of 125°C.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

SPECIFICATION

MILITARY

MIL-R-23648 - Resistor, Thermal (Thermistor), Insulated, General Specification For.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Building 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center, Columbus, ATTN: DSCC-VAM, Post Office Box 3990, Columbus, OH 43216-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5905
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3. REQUIREMENTS

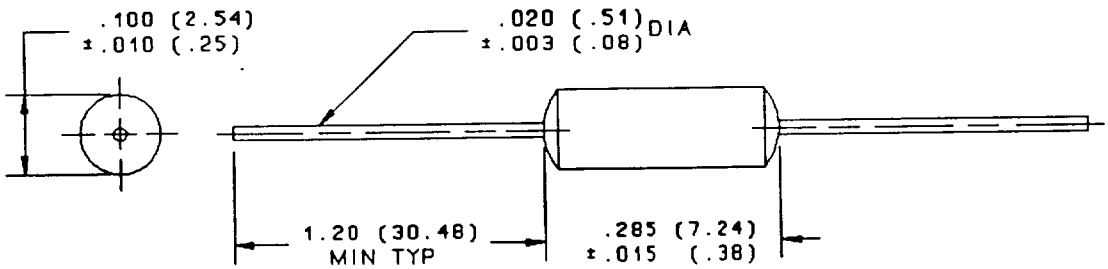
* 3.1 Requirements. Requirements shall be in accordance with MIL-R-23648, and as specified herein.

3.2 Design and construction. The thermistors shall be of the design, construction, and physical dimensions specified on figure 1.

3.3 Thermal time constant. The thermal time constant shall be 60 seconds maximum.

3.4 Dissipation constant. The dissipation constant shall be 2.5 milliwatts per degrees Celsius minimum.

3.5 Power rating. The thermistor shall be capable of dissipating a maximum power of 0.25 watts at 25°C. Thermistors shall be derated in accordance with figure 2.



NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents are given for general information only.

FIGURE 1. Style RTH42 thermistors.

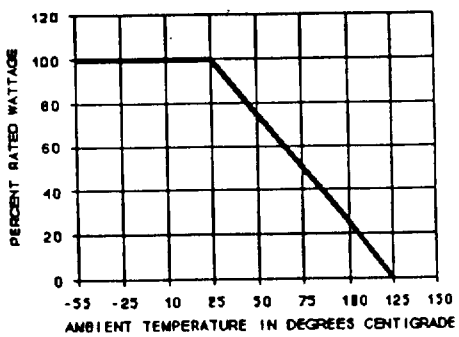


FIGURE 2. Derating curve for high ambient temperatures.

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3.6 Resistance.

* 3.6.1 Resistance at 25°C. Standard values shall be as specified in MIL-R-23648 except that the 5 percent decade values shall also be used for resistance tolerance G. Minimum and maximum resistance values shall be in accordance with table I.

TABLE I. Minimum and maximum resistance values (at 25°C).

Ratio 0.55 (E)
ohms
10 min 10k max

3.6.2 Resistance at temperatures other than 25°C. Factors for determining resistance at temperatures other than 25°C are in accordance with table II. The appropriate factor is selected from the column headed by the resistance range which includes the zero-power 25°C resistance of the thermistor in question. The 25°C resistance of the thermistor is multiplied by the factor selected to obtain the resistance at any given temperature.

Example: Given a thermistor with a 25°C resistance of 220 ohms, find the resistance at 75°C. Select the factor opposite 75°C from the column headed by the resistance range containing 220 ohms. The factor 1.400 is thus selected from the column headed 180-470. Multiply 220 ohms by the factor 1.400 to obtain the resistance at 75°C of 308 ohms.

TABLE II. Factors for determing resistance at various temperatures.

	Resistance range (ohms)					
Temperature (°C)	10-68	82-150	180-470	560-1.2k	1.5k-5.6k	6.8k-10k
-55	.615	.582	.560	.550	.515	.510
-15	.790	.770	.755	.740	.730	.730
0	.863	.847	.838	.835	.825	.825
25	1.0	1.0	1.0	1.0	1.0	1.0
50	1.160	1.170	1.180	1.20	1.230	1.190
75	1.350	1.370	1.400	1.420	1.450	1.400
100	1.545	1.584	1.623	1.656	1.670	1.610
125	1.750	1.80	1.860	1.920	1.960	1.830

3.7 Short time overload. The maximum allowable change in zero-power resistance as the result of the short time load test shall be 2 percent.

3.8 Low temperature storage. The maximum allowable change in zero-power resistance as the result of the low temperature storage test shall be 2 percent.

3.9 High temperature storage. The maximum allowable change in zero-power resistance as the result of the high temperature storage test shall be 1 percent.

3.10 Terminal strength. When tested in accordance with 4.2, the maximum allowable change in zero-power resistance shall be 1 percent.

3.11 Thermal shock. The maximum allowable change in zero-power resistance as the result of the thermal shock test shall be 2 percent.

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3.12 Resistance to soldering heat, type S. The maximum allowable change in zero-power resistance as the result of the resistance to soldering heat test shall be 1 percent.

3.13 Moisture resistance. The maximum allowable change in zero-power resistance as the result of the moisture resistance test shall be 3 percent.

3.14 Life. The maximum allowable change in zero-power resistance as the result of the life test shall be 5 percent.

3.15 High temperature exposure. The maximum allowable change in zero-power resistance as the result of the high temperature exposure test shall be 1 percent after 100 hours, and 2 percent after 1,000 hours.

3.16 Vibration, high frequency. The maximum allowable change in zero-power resistance as the result of the vibration test shall be 2 percent.

3.17 Shock, specified pulse. The maximum allowable change in zero-power resistance as the result of the shock test shall be 2 percent.

3.18 Seal. When tested in accordance with 4.5, the maximum allowable leakage rate shall be 1×10^{-7} atmospheric cubic centimeters per second (atm cc/sec). for the purpose of this specification, a hermetically sealed thermistor is one which is capable of passing the seal test of 4.5. Enclosure materials shall be ceramic, metal, or glass, or combinations thereof.

3.19 Immersion. The maximum allowable change in zero-power resistance as the result of the immersion test shall be 3 percent.

* 3.20 Resistance to solvents. There shall be no evidence of mechanical damage to the body and the marking shall remain clear and legible.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-23648.

* 4.1.1 Qualification seal test. The seal test specified in 4.5 shall be performed prior to the immersion testing of MIL-R-23648, table III, group VI.

* 4.1.2 Semiannual seal test. The seal test specified in 4.5 shall be performed prior to the immersion test of MIL-R-23648, table VI.

4.2 Terminal strength.

4.2.1 Direct load. Direct load shall be applied gradually until the load reaches 4.5 pounds.

4.3 Solderability (type S leads). The solderability test is applicable to this specification for the type S leads.

4.4 Resistance to soldering heat (type S leads). The resistance to soldering heat test is applicable to this specification.

4.5 Seal. Thermistors shall be tested in accordance with method 112 of MIL-STD-202. The following details shall apply.

a. Test condition: C, procedure III.

b. After test: Thermistors shall be cleaned in alcohol or other suitable degreaser, and allowed to dry thoroughly.

5. PACKAGING

* 5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-23648.

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6. NOTES

* 6.1 Notes. In addition to the notes specified herein, the notes specified in MIL-R-23648 are applicable.

* 6.2 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, AT, AV, ME
Navy - AS, CG, MC, OS
Air Force - 17, 19, 99

Preparing activity:
DLA - ES

(Project 5905-1411-08)

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